Local
Intrinsic Global
Environmental Global

Random Forest

XGBoost

All Centrals

Relative Importance

Random Forest: Train AUC = 0.83, Test AUC = 0.82
Local 17%, Env. (4%), Intr. 79%

XGBoost: Train AUC = 0.84, Test AUC = 0.82
Local 21%, Env. 16%, Intr. 63%

M_{BH} Σ^* r/R_e M^* δ_5 M_h

High-Mass Satellites

Random Forest: Train AUC = 0.77, Test AUC = 0.74
Env. (0%)
Local 18%, Intr. 82%

XGBoost: Train AUC = 0.76, Test AUC = 0.73
Local 28%, Env. 15%, Intr. 57%

M_{BH} Σ^* M^* r/R_e M_h δ_5

Low-Mass Satellites

Random Forest: Train AUC = 0.79, Test AUC = 0.80
Local (0%)
Intr. 8%, Env. 92%

XGBoost: Train AUC = 0.84, Test AUC = 0.82
Local 8%, Intr. 32%, Env. 60%

M_h M_{BH} M^* δ_5 Σ^* r/R_e